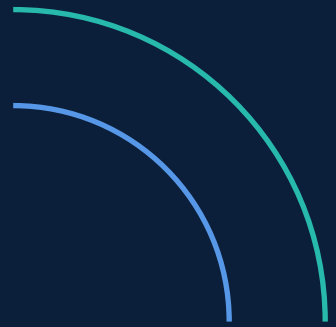
MICROSOFT FILE SYSTEMS

Disk Partitions

Exploring Microsoft file structures for Windows/DOS storage and forensic analysis



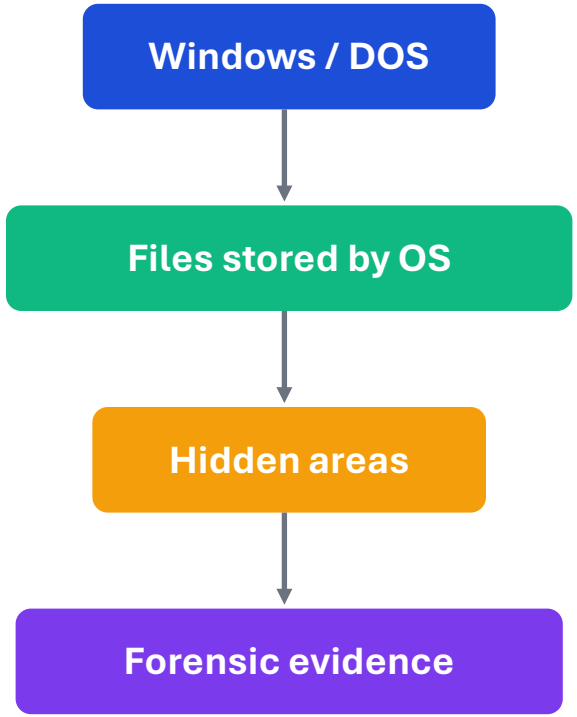


Context

Why Microsoft file structures matter

The OS storage method determines where data can be stored — and where it can be hidden.

Most PCs use Microsoft software products, so Windows/DOS file systems are important to understand.
Core ideas: clusters, FAT, NTFS, partitions, boot records and file-structure databases.
In forensics, hidden areas may contain full files or fragments of files that become evidence.

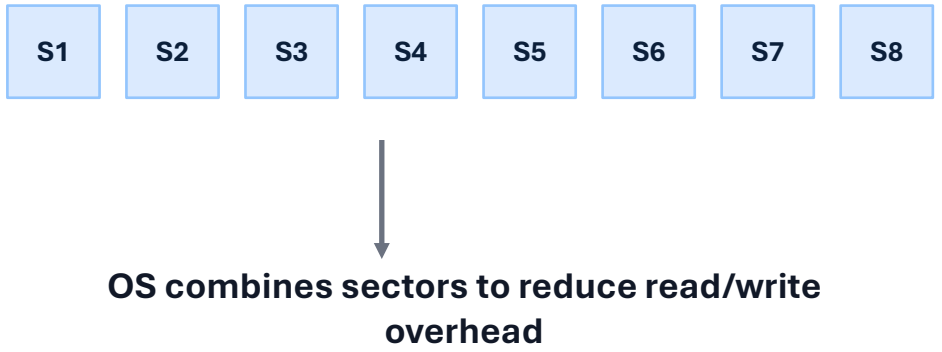


Storage structure is not just theory; it decides what investigators can find.

Sectors grouped into clusters

Sector

A small physical storage unit on the disk.



Cluster

One or more sectors treated as one allocation unit by the OS.



Typical cluster sizes: 512, 1024, 2048, 4096 bytes or more

Cluster addresses vs sector addresses

Logical address

Cluster numbers assigned by the OS



Clusters start from 2

Example from textbook

```
Cluster 100
= 98 clusters from
Cluster 2
```

Physical address

Sector numbers at hardware/firmware level



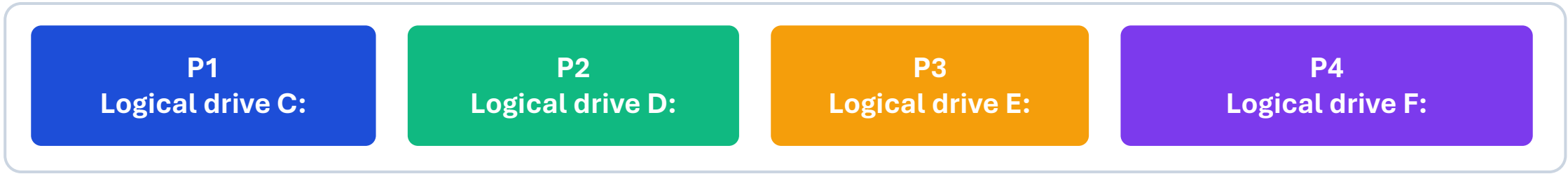
Sectors start from 0

Clusters and their addresses are specific to a logical disk drive, which is a disk partition.

Disk partitions create logical drives

A partition is a section of a physical disk treated as a logical drive.

Physical hard disk



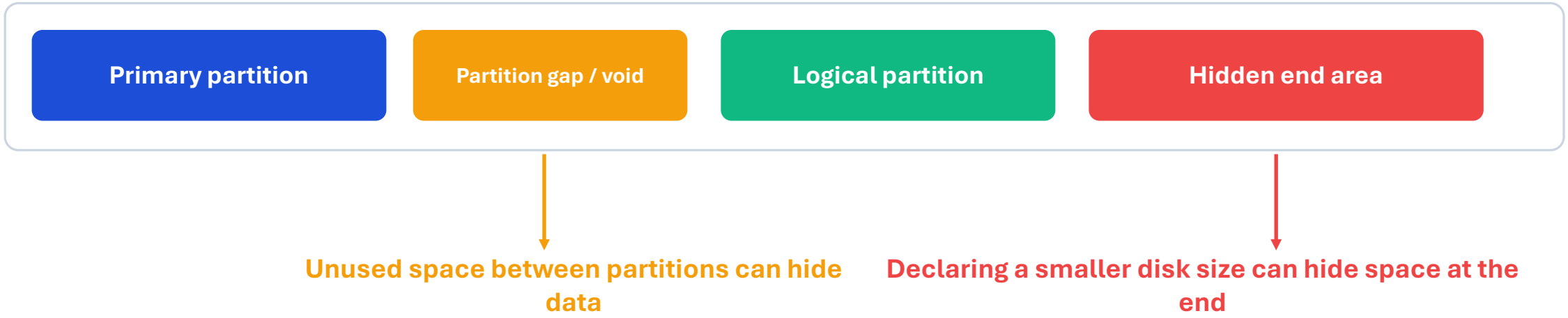
Example: an 8 GB hard disk can contain four partitions or logical drives.

Older file systems had size limitations, so partitioning helped the OS recognize available space.

Forensic focus

Where data can be hidden

Partition layout with visible and hidden areas



Hidden partition: reference removed or concealed from the partition table.
Void: large unused gap between partitions.
Partition gap: unused space between the primary partition and first logical partition.
Disk-editing tools can access these hidden or empty areas.

Tools

Examining partitions at the physical level

Disk editors show the raw disk contents that normal file browsing hides.

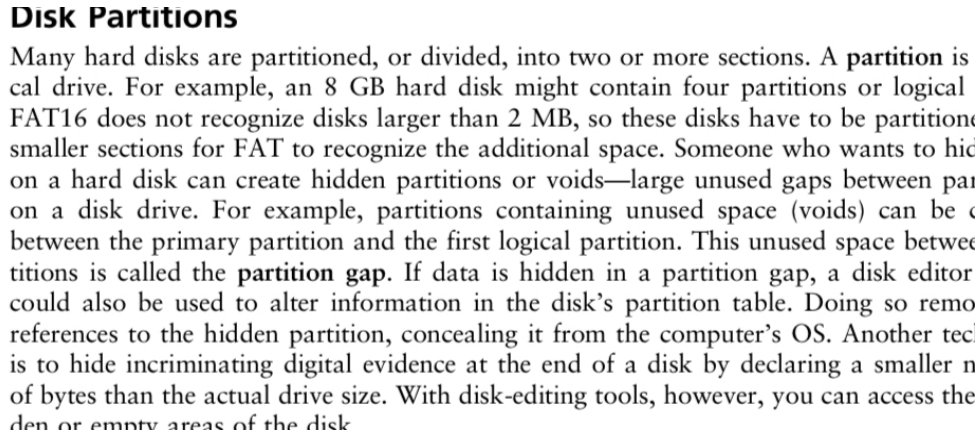
Norton DiskEdit

WinHex

Hex Workshop

Used to examine

File headers
Boot records
Partition tables
Critical parts of files
Hexadecimal codes used by the OS



Disk Partitions

Many hard disks are partitioned, or divided, into two or more sections. A **partition** is
cal drive. For example, an 8 GB hard disk might contain four partitions or logical
FAT16 does not recognize disks larger than 2 MB, so these disks have to be partitione
smaller sections for FAT to recognize the additional space. Someone who wants to hid
on a hard disk can create hidden partitions or voids—large unused gaps between par
on a disk drive. For example, partitions containing unused space (voids) can be c
between the primary partition and the first logical partition. This unused space betwee
titions is called the **partition gap**. If data is hidden in a partition gap, a disk editor
could also be used to alter information in the disk's partition table. Doing so remo
references to the hidden partition, concealing it from the computer's OS. Another tec
is to hide incriminating digital evidence at the end of a disk by declaring a smaller n
of bytes than the actual drive size. With disk-editing tools, however, you can access the
den or empty areas of the disk

Textbook reference excerpt: disk partitions and disk-editing tools

Table 6-1

Partition table hexadecimal codes — Microsoft and hidden types

Hex code	File system / partition type
01	DOS 12-bit FAT
04	DOS 16-bit FAT for partitions smaller than 32 MB
05	Extended partition
06	DOS 16-bit FAT for partitions larger than 32 MB
07	NTFS
0B	DOS 32-bit FAT
0C	DOS 32-bit FAT for interrupt 13 support
17	Hidden NTFS partition (XP and earlier)
1B	Hidden FAT32 partition
1E	Hidden VFAT partition
3C	Partition Magic recovery partition

dentify and maintain the file system. Table 6-1 lists the hexadecimal codes in a p
able and identifies some common file system structures.

Table 6-1 Hexadecimal codes in the partition table

Hexadecimal code	File system
01	DOS 12-bit FAT
04	DOS 16-bit FAT for partitions smaller than 32 MB
05	Extended partition
06	DOS 16-bit FAT for partitions larger than 32 M
07	NTFS
08	AIX bootable partition
09	AIX data partition
0B	DOS 32-bit FAT
0C	DOS 32-bit FAT for interrupt 13 support
17	Hidden NTFS partition (XP and earlier)
1B	Hidden FAT32 partition
1E	Hidden VFAT partition
3C	Partition Magic recovery partition
66–69	Novell partitions
81	Linux

Textbook table excerpt

Codes reveal the partition/file-system type, including hidden partitions.

Table 6-1 continued

Partition table hexadecimal codes — other systems

Hex code	File system / partition type
08	AIX bootable partition
09	AIX data partition
66–69	Novell partitions
81	Linux
82	Linux swap partition; may also be associated with Solaris partitions
83	Linux native file systems: Ext2, Ext3, Reiser, Xiafs
86	FAT16 volume/stripe set (Windows NT)
87	HPFS mirrored partition or NTFS volume/stripe set
A5	FreeBSD and BSD/386
A6	OpenBSD
A9	NetBSD
C7	Typical corrupted NTFS volume/stripe set
EB	BeOS

07 = NTFS

82 = Linux swap

C7 = corrupted NTFS

Procedure

Identifying the file system using Hex Workshop

1. Install Hex Workshop if needed, as directed by the instructor.
2. Insert a USB drive into a USB port.
3. Run Hex Workshop as administrator.
4. Choose Disk → Open Drive, select C: or working drive, click OK.
5. Open the USB drive and compare its file-system label.

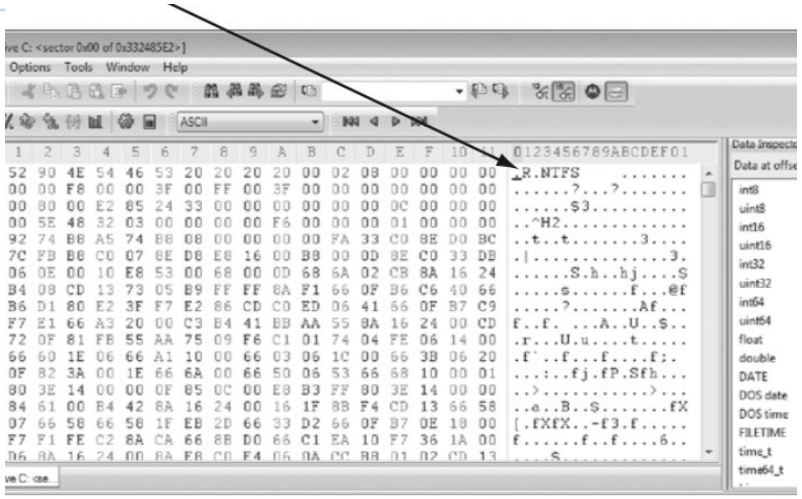


Figure 6-4 reference: first logical sector shows file-system label

```
NTFS: ".R.NTFS"
FAT: "MSDOS5.0" or "MSWIN4.1"
```

NTFS and FAT labels in the first logical sector

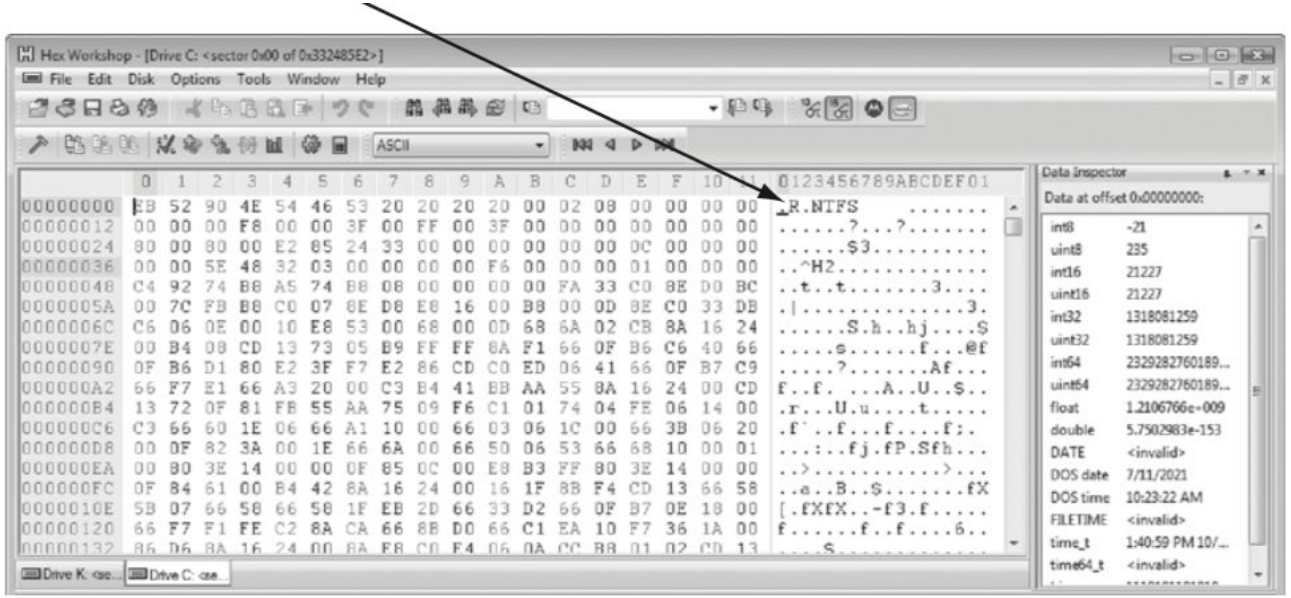


Figure 6-4: Hex Workshop identifying the file system

What to look for

`.R.NTFS`

Displayed when the partition is formatted as NTFS.

`MSDOS5.0`
`MSWIN4.1`

Displayed in the first logical sector for FAT drives.

Useful for identifying a drive even before browsing normal files.

File headers: identifying a file without its extension

A file header is a byte pattern at the beginning of a file that identifies its real type.

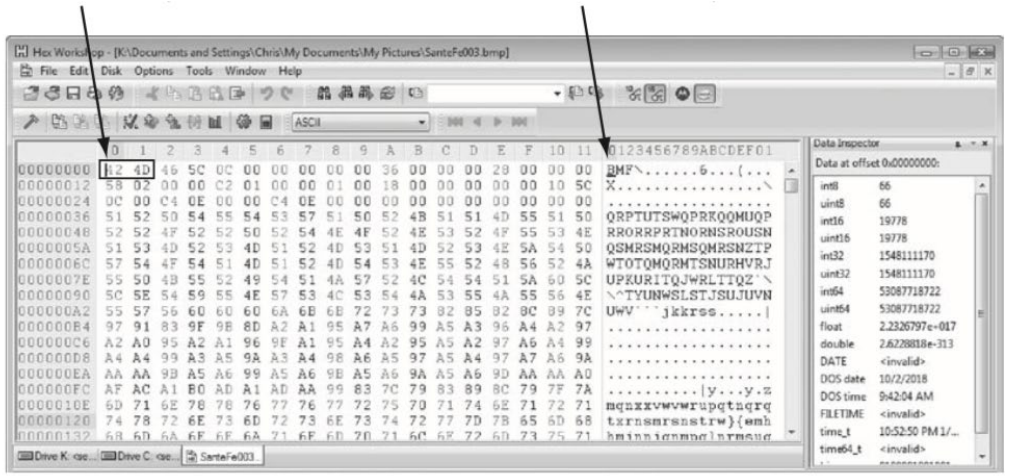


Figure 6-5 Hex Workshop indicating a .bmp file

Figure 6-5: Hex Workshop indicating a .bmp file

Bitmap file markers

BM6 BM BMF

42 4D = BM

42 = hexadecimal “B”
4D = hexadecimal “M”
A changed or missing extension can still be detected from the header.

Microsoft Office file header

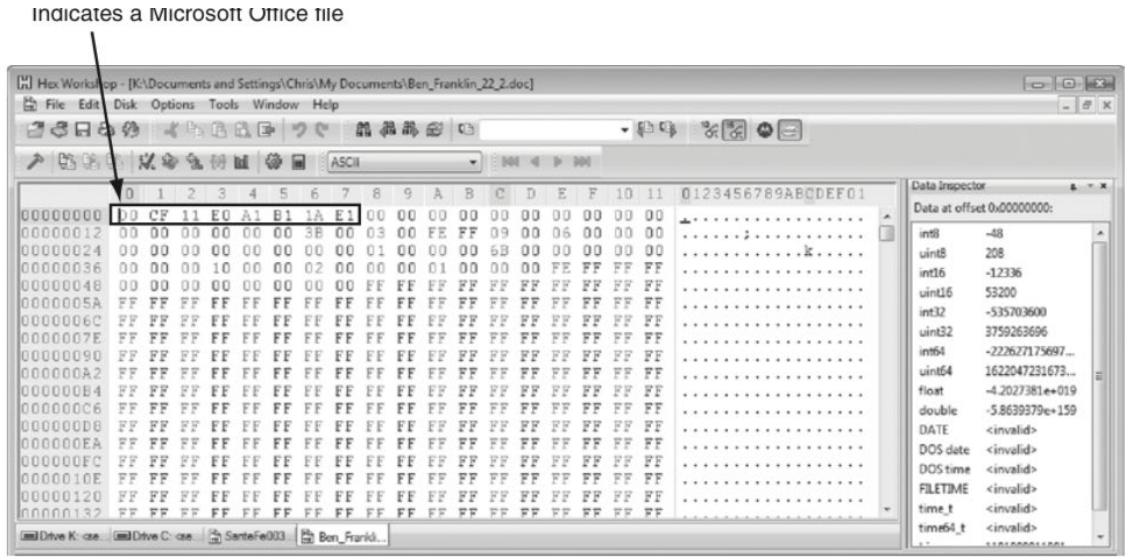


Figure 6-6 Hex Workshop indicating a Microsoft Office file

Figure 6-6: Hex Workshop indicating a Microsoft Office file

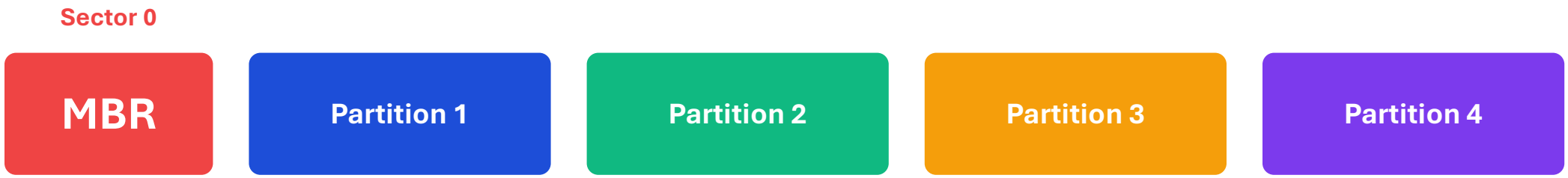
Classic Office signature

```
D0 CF 11 E0
A1 B1 1A E1
```

Identifies older Microsoft Office documents.
Can appear in Word, Excel or PowerPoint files.
Textbook note: does not apply to Access databases.
Hex values may be grouped in sets of two or four digits.

Master Boot Record (MBR)

The MBR stores partition information needed by traditional Windows/DOS boot disks.



Stores:

Partition locations
Partition sizes
Other boot-related information

Boot utilities can modify the MBR, so forensic acquisition tools may be affected.

Practical point: keep multiple data-acquisition tools available.